

Multiple Sclerosis (MS)

Multiple Sclerosis (MS) is a chronic disease that affects the central nervous system, namely the brain and spinal cord, and occurs when the immune system mistakenly attacks its own nerve tissues. The main structure targeted by the immune system is the myelin sheath surrounding the nerves. Myelin helps ensure that communication between nerves occurs properly. Damage to this structure can disrupt communication between the brain and other parts of the body. Therefore, MS can manifest with different symptoms from person to person.

MS is particularly common in young adults and is more frequently seen in women than in men. It is estimated that more than 1.8 million people worldwide are living with MS. The course of the disease is not the same for everyone. In some people, symptoms may completely or partially improve after they appear, while in others, symptoms may become permanent over time. Therefore, it is not correct to consider MS as a disease that progresses in only one way.

Damage to the myelin sheath can disrupt the transmission of messages sent through the nerves. As a result, symptoms such as vision problems, numbness, muscle weakness, balance and walking problems, muscle stiffness, and fatigue may occur. In addition, problems with bladder control, dizziness, speech difficulties, depression, and changes in cognitive functions such as attention, memory, or thinking may also be seen. However, an important point is that not all of these symptoms occur in every person with MS. Symptoms may vary depending on which area of the brain or spinal cord is affected.

One of the most common forms of MS is relapsing-remitting MS (RRMS). In this form, new symptoms may appear during certain periods, followed by periods in which the symptoms completely or partially improve. In some people, however, the disease may show more continuous progression over time. In primary progressive MS, symptoms gradually worsen from the beginning, whereas in secondary progressive MS, a disease that initially follows a relapsing course may later become continuously progressive.

The exact cause of MS is not yet known. However, genetic and environmental factors are thought to play a role together. Having a family history of MS may increase susceptibility to the disease, but MS is not considered a directly hereditary disease. Research has found associations between MS and various factors such as the Epstein-Barr virus, sunlight, and vitamin D levels. Smoking has also been reported to be associated with an increased likelihood of developing MS and with a more severe course of the disease.

There is no single test that can definitively identify MS. For diagnosis, the person's symptoms and neurological examination are evaluated; in addition, magnetic resonance imaging (MRI) is particularly used. Lesions associated with MS may be observed on MRI scans. When necessary, other tests such as examination of the cerebrospinal fluid, evoked potential tests, and optical coherence tomography may also be used.

There is currently no definitive cure for MS. However, effective treatments are available to control the disease. The main goals of treatment are to reduce relapses, slow disease progression, and control symptoms. Corticosteroids may be used during relapse periods. Disease-modifying treatments can help reduce new relapses and disease activity. In addition, physical therapy and rehabilitation can support the preservation of mobility and help make daily life easier.

Another noteworthy aspect of MS is that, in some people, the disease may be detected on MRI scans before symptoms become apparent. In this condition, called radiologically isolated syndrome (RIS), a person may have MRI findings suggestive of MS without yet having clear MS symptoms. A large study published in 2025 showed that approximately one-third of 273 individuals diagnosed with RIS developed MS symptoms during follow-up and that some biomarkers may help predict this risk.

These findings are important for recognizing the disease at an early stage and for more closely monitoring individuals at risk.

To sum up, MS is a disease that occurs when the immune system damages the myelin in the central nervous system and can vary from person to person. It can cause a wide range of symptoms, from vision problems and fatigue to numbness and balance problems. The disease does not progress in the same way in every person with MS. Although there is currently no definitive cure, early diagnosis, appropriate treatment, and rehabilitation can help control the disease and preserve quality of life.

References:

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